

Geneva Ecola

gecola@stanford.edu | genevaecola.com | linkedin.com/in/geneva-ecola

Education

- Expected 2027** **Ph.D. Electrical Engineering, Stanford University**
Thesis: Harnessing Ambient Radio Signals for Ultra-Low Power Connectivity
Advisor: Zerina Kapetanovic, *Thesis Committee:* Howard Zebker, Joseph Kahn
- 2023** **M.S. Electrical Engineering, Stanford University**
- 2019** **B.S. Engineering, Harvey Mudd College**

Industry Experience

- Summer 2024** **Radio Frequency Intern, Apple**
Developed new testbed to analyze the impact of chip packaging variants on radio receiver performance. Automated optical chip-fiber alignment and measured photonic structures.
- Summer 2023** **Wireless Hardware Intern, Tesla**
Evaluated localization technologies for a passive entry system including BLE and UWB. Designed and evaluated RF balun circuit matching between chip and antenna.
- 2019 – 2021** **Senior Radio Frequency Engineer, Trellisware Technologies**
Redesigned the high-frequency analog front-end of radios using simulations and tested hardware in lab. Led hardware sustainment of a software-defined radio digital board, coordinating team of 10+ engineers.
- Summer 2018** **Antenna Intern, Echodyne**
Measured electronically scanning array antenna near field at multiple scan angles and frequencies and developed MATLAB scripts to quantify antenna performance metrics.

Publications

(*indicates co-first authorship)

- 2026** **First-Principles Channel Model and Capacity-Achieving Transmission Scheme in Thermal Noise Backscatter Links**
Geneva Ecola*, Jida Zhang*, Joesph Kahn and Zerina Kapetanovic
Under Review
- 2025** **Wireless Johnson Noise Thermometry for Passive Temperature Sensing**
Geneva Ecola*, Jasmin Falconer* and Zerina Kapetanovic
IEEE Journal of Microwaves
- 2025** **SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar**
Geneva Ecola, Bill Yen, Ana Banzer Morgado, Bodhi Priyantha, Ranveer Chandra, and Zerina Kapetanovic
ACM SenSys

Honors

- 2026** **Best Presentation at ACM MobiSys Rising Star Forum**
- 2026** **Best Poster Runner Up at ACM MobiSys N2Women Workshop**
- 2026** **Stanford Dissertation Completion Fellowship**
- 2026** **Qualcomm Innovation Fellowship Finalist**
Proposal: Ultra-low Energy IoT Networks Using Thermal Noise Communications
- 2025** **Stanford Emerging Environmental Scholar Research Award**
Proposal: Bringing Smart Sensing Systems Everywhere on Earth
- 2024** **Stanford Graduate Fellowship, Rambus Corporation Fellow**
- 2024** **Apple-Stanford Integrated Systems PhD Fellowship**
Proposal: Enabling Ultra-Low Power Smart Sensing Systems
- 2023** **Stanford EE315 Best Analog to Digital Converter Design**
- 2021** **Stanford Graduate Fellowship, Rambus Corporation Fellow**
- 2019** **Harvey Mudd College Departmental Honors in Engineering**
- 2018** **Harvey Mudd College T. Larry Norin Memorial Scholarship**
- 2018** **Harvey Mudd College Best Engineering Clinic Presentation**
- 2015** **Harvey S. Mudd Merit Award**

Talks

- 2026** **Harnessing Ambient Radio Signals for Ultra Low Power Sensing and Communication Systems**
ACM MobiSys Rising Star Forum
- 2026** **Ultra-low Energy IoT Networks Using Thermal Noise Communications**
Apple Fellowship Talk
- 2025** **Doing More with Less: Harnessing Ambient Signals for Wireless Sensing and Communication**
University of Washington Department of Electrical Engineering
Apple Fellowship Talk
- 2025** **SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar**
ACM SenSys
Apple Fellowship Talk

Posters

- Ultra-low Energy IoT Networks Using Thermal Noise Communications**
Geneva Ecola, Jida Zhang and Zerina Kapetanovic
2026 *ACM MobiSys N2Women Workshop*
2026 *Qualcomm Innovation Fellowship Finals*
- Vision Assisted Beamsteering for Full Control of Wireless Power Transfer and Joint Communication Link**
Jasmin Falconer, Geneva Ecola, and Zerina Kapetanovic
2025 *Stanford Center for Image Systems Engineering Affiliates Annual Meeting*
- SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar**
Geneva Ecola, Bill Yen, Ana Banzer Morgado, Bodhi Priyantha, Ranveer Chandra, and Zerina Kapetanovic
2025 *ENSsys Workshop: Energy Harvesting & Energy-Neutral Sensing Systems*
2024 *Stanford Center for Image Systems Engineering Affiliates Annual Meeting*

Teaching Experience

- 2026** **Course Development, Internet of Things, Stanford University**
Developed labs and final project for Internet of Things course to teach students about building devices and enabling internet connectivity via different wireless protocols.
- 2025** **Course Assistant, Circuits I, Stanford University**
Ran lab sections and office hours for 30+ students in introductory circuits course. Gave midterm and final review lectures.
- 2023** **Course Development and Assistant, 3D+ Imaging Sensors, Stanford University**
Led lab sessions and office hours for 30+ students on ultrasonic sensors, time-of-flight cameras, and mmWave radar. Designed final project to design a through-wall imaging system using mmWave radar.
- 2017 – 2018** **Teaching Assistant, Systems Engineering, Harvey Mudd College**
Taught concepts needed to complete lab experiments and provided debugging help.

Advising

- 2025 – Present** Rudraksh Mohapatra, Stanford University Undergraduate
2024 Sylvia Chen, Stanford University Undergraduate

Outreach

- 2024 – 2025** **Stanford Women in Electrical Engineering**
Vice President
Organized professional development events and community building, including a 50+ person mentorship program.
- 2023** *Outreach Chair*
Hosted outreach events including teaching high schoolers at Stanford Splash.

Service

	Reviewer
2025, 2026	<i>IEEE Journal of Microwaves</i>
2025	<i>ACM SenSys Poster PC Member</i>
2026–Present	Branch Chair and Founder, Stanford IEEE Microwave Theory & Technology Host workshops and talks on microwave engineering topics to enable networking opportunities and create a community of student interested in wireless design.
2024–2025	Student Life and Culture Committee Stanford University Department of Electrical Engineering